DO NOW — RECALL

Lesson 146 · Network Threats: Attack Paths and Conseq

Name: _____ Class: _____ Date: _____

1. When packets arrive out of order, what can TCP do?

2. Which layer in TCP/IP handles IP routing?

3. What data structure follows FIFO?

4. What is the result of an AND gate with inputs 1 and 0?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Use sequence information to reassemble the byte stream correctly.

2. Model answer

The internet layer.

3. Model answer

A queue.

4. Model answer

0.

By the end, you can…

• Distinguish malware, phishing, interception and denial-of-service threats without conflating them.

• Trace how a network attack reaches an asset and identify the vulnerability involved.

• Explain an appropriate defence for the particular threat rather than naming generic 'security'.

Vocabulary — know these terms

phishing — Social engineering to obtain information or actions through deception.

malware — Software designed to perform harmful or unauthorised actions.

DDoS — Distributed denial-of-service intended to disrupt availability.

CIA triad — Confidentiality, integrity and availability.

NEW LEARNING

Read, annotate and explain

Entry point

Suspicious link, exposed service or weak account.

Threat action

Credential theft, malware execution or traffic flooding.

Effect on service

Compromised confidentiality, integrity or availability.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A pupil receives a fake school login link and submits a password. Explain the attack.

Fake login email

Identify the facts and make an initial prediction.

Credentials submitted

Explain the mechanism or trace the intermediate stage.

Account access at risk

Phishing tricks the pupil into visiting an attacker-controlled form. The secret is disclosed, then the attacker may attempt unauthorised account access. Awareness and phishing-resistant MFA help, while a firewall alone does not prevent users entering secrets into fake sites.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain how phishing differs from malware and give one example of each. [6]

Question 2

A school site is reachable but pupils' account details are exposed. Identify the affected security property and two controls. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A network suffers DDoS and suspected password theft simultaneously. Prioritise two distinct mitigations and justify why a single firewall rule is inadequate. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Encrypting a site with HTTPS is sufficient to prevent a DDoS attack.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.